Blink Order System

Store order and inventory system: project overview, flow and API reference

Laravel 8 | PHP 7.4 | MySQL | REST API | Queued jobs

1. Features

- Normalized MySQL schema, built with migrations and Eloquent relationships
- REST APIs for customers, order creation, order history and low-stock products
- Automatic subtotal, tax and grand-total calculation
- Stock deducted automatically when an order is placed
- Database transactions with row-level locking, so stock cannot be oversold
- Order-confirmation emails sent by a queued job
- Request validation and consistent JSON error responses
- Factories and seeders for sample products and customers
- Web pages for orders, customers, products, low stock and restocking

2. Database

```
customers ──< orders ──< order_items >── products
```

Table	Key columns	Rules
`products`	name, code (unique), price, tax_percentage, stock_on_hand	Stock cannot go below 0
`customers`	name, email (unique)	
`orders`	customer_id, subtotal, tax_total, grand_total	A customer with orders cannot be deleted
`order_items`	order_id, product_id, quantity, unit_price, tax_percentage, subtotal, tax_amount, total	Price and tax are copied from the product when the order is placed. A product used in an order cannot be deleted
`jobs`, `failed_jobs`		Used by the queue

Relationships: `Customer → orders`, `Order → customer, items`, `OrderItem → order, product`, `Product → orderItems`.

3. Order flow

```
POST /api/orders  or  the web "Place order" form
  │
  ├─ StoreOrderRequest validates: customer exists, at least 1 item,
  │  each product once, quantity ≥ 1
  ▼
OrderService::create()
  ├─ DB transaction
  │   ├─ lock the product rows (SELECT … FOR UPDATE)
  │   ├─ check stock → if short: 422 for the API, form error on the web
  │   ├─ create the order
  │   ├─ for each item: copy price and tax, calculate the line, reduce stock
  │   └─ save subtotal, tax_total and grand_total
  ├─ commit
  └─ SendOrderConfirmation::dispatch($order)        → queued job
        ▼
     Mail::to(customer email) → OrderConfirmationMail
     (plain text, written to storage/logs/laravel.log)
```

Money

- Calculations use `bcmath`, so there are no floating-point rounding errors.
- Line subtotal = unit price × quantity. Line tax = subtotal × tax %, rounded half up.
- The order totals are the sum of the rounded lines, so they always match.
- Amounts are shown with ₹ and Indian digit grouping, for example ₹1,23,456.78.

Queue

- The email job is queued only after the order's transaction commits.
- A failing job is tried 3 times, waiting 10 seconds and then 60 seconds, then moves to `failed_jobs`.
- `QUEUE_CONNECTION=sync` runs the job immediately; `database` needs `php artisan queue:work`.

Restocking

Stock is added in a single atomic database update, so it is safe while orders are being placed. Each restock is logged. Product edit forms cannot overwrite the stock count.

4. API

Base URL: `http://127.0.0.1:8000/api` · Headers: `Accept: application/json`, `Content-Type: application/json`
Postman collection: `postman/blink-order-system.postman_collection.json`

Customers

Method	Endpoint	Body or query
GET	`/customers`	`?search=&per_page=`
POST	`/customers`	`{"name": "Arun Kumar", "email": "arun.kumar@example.com"}`
GET	`/customers/{id}`	
PATCH	`/customers/{id}`	`{"name": "..."}`
DELETE	`/customers/{id}`	204, or 409 if the customer has orders
GET	`/customers/{id}/orders`	Customer's order history

Products

Method	Endpoint	Body or query
GET	`/products`	`?search=&status=in_stock|low_stock|out_of_stock`
GET	`/products/low-stock`	`?threshold=` (default 10)
GET	`/products/{id}`	
PATCH	`/products/{id}`	`{"name": "...", "price": "46.50", "tax_percentage": "6"}`
POST	`/products/{id}/restock`	`{"quantity": 50}`

Orders

Method	Endpoint	Body or query
POST	`/orders`	See payload below
GET	`/orders`	`?customer_id=&from=YYYY-MM-DD&to=YYYY-MM-DD`
GET	`/orders/{id}`	

```
{
  "customer_id": 5,
  "items": [
    { "product_id": 7, "quantity": 2 },
    { "product_id": 8, "quantity": 1 }
  ]
}
```

Prices are never accepted from the client. They are read from the products table when the order is placed.

Money in responses

```
"currency": "INR",
"grand_total": "110.56",
"grand_total_formatted": "₹110.56"
```

Errors

Status	When
422	Validation failed, or not enough stock (lists requested and available quantity for each product)
404	`{"message": "Resource not found."}`
409	Deleting a customer who has orders

5. Web pages

URL	Page
`/orders`, `/orders/create`, `/orders/{id}`	Order history with filters, place-order form, order detail
`/customers` (+ create, edit, show)	Customer list with search; the detail page shows order history
`/products`	Catalogue with search, stock-status filter, and restock or edit on each row
`/products/low-stock`	Low-stock list with an adjustable threshold and an Add button
`/products/{id}/edit`	Add stock, and edit name, code, price and tax

6. Code layout

Layer	Files
Models	Customer, Product, Order, OrderItem
API controllers	Api/CustomerController, CustomerOrderController, OrderController, ProductController
Web controllers	Web/CustomerController, OrderController, ProductController
Form requests	Customer/*, Order/*, Product/*, Web/PlaceOrderRequest
API resources	CustomerResource, ProductResource, OrderResource, OrderItemResource
Services	OrderService (orders, stock, totals), InventoryService (restocking)
Queue and mail	Jobs/SendOrderConfirmation, Mail/OrderConfirmationMail, emails/orders/confirmation.blade.php
Errors	InsufficientStockException, Handler, ForceJsonResponse middleware
Helpers	Support/Money (₹ formatting), `@money` Blade directive
Data	CustomerFactory, ProductFactory, ProductSeeder, CustomerSeeder
Postman	postman/blink-order-system.postman_collection.json

7. Setup

```
composer install
cp .env.example .env
php artisan key:generate
php artisan migrate --seed
php artisan serve                 # http://127.0.0.1:8000
php artisan queue:work            # only if QUEUE_CONNECTION=database
```

8. Sample data

Demo customer: `demo.customer@example.com`

Code	Product	Starting stock
COF-001	Arabica Coffee Beans 1kg	120
TEA-001	Green Tea 50 Bags	80
CUP-008	Paper Cup 8oz (100 pcs)	8 (low)
MLK-OAT	Oat Milk 1L	3 (low)
SYR-CHO	Chocolate Syrup 750ml	0 (out of stock)

The seeders also add 18 random products and 10 random customers.